

PEDRO HENRIQUE ZARRANZ LOURES · UNITY DEVELOPER · TORONTO

Pedro H. Z. Loures

I turn game designs into systems that ship.

Unity developer since 2012: gameplay systems, VR interaction, editor and Blender tools, networking and data pipelines, for studios, schools and clients like Fiat and Vale. A Vale presentation system and a Blender animation add-on I built are still in use. Most recently I was Unity lead on Anchored Memories, a full-body haptic VR experience at UT Austin.

Unity · C#

Gameplay systems

Tools · Python

Networking & data

VR / Haptics

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SELECTED PROJECTS



HAPTIC VR · UNITY · 2026

Anchored Memories

Five-act haptic VR story on climate migration · Unity lead and project engineer · UT Austin, 2026

Role Unity Lead · Project Engineer (independent partner, volunteer) · **Year** 2026 · **Engine** Unity ·

Team 12+ (TXI, D2R Studios, independent partners, UT Austin advisors) · **For** Texas Immersive Institute, UT Austin

- Built a node-based story graph on Unity's graph tooling, back when it wasn't yet built into the editor, so students could sequence story beats and VR interactions, with cues and timed hints, without writing code
- Engineered the project's foundation: the barebones architecture plus the base code for VR interactions and audio that the team built on
- Set up the structure of all five acts as separate spaces, each with its own environment prepared for the ocean changes, ready for the team's level design
- Helped plug the haptics designer's bHaptics SDK work into the Unity project
- Built shaders (including the sand) and rain particle effects with optimization tests; imported and configured assets, including the ocean, lent from my own Asset Store library where licenses allowed
- Helped test the release build on my own VR headset

LEADERSHIP

- Moved the project from Unreal to Unity when I joined, to lower the deadline risk on the program's first big project with outside partners and reach the July 2026 showcase
- Main developer on the levels for part of the project, and on call as backup throughout the year
- Part of the planning team: kept the GitHub project board in order, helped divide tasks around weekly check-ins, and onboarded the students into the project



MANAGEMENT SIM · UNITY · 2021

Dream Arcade Simulator

Arcade management sim · Centennial College capstone, team of five

Role Gameplay Programmer · **Year** 2021 · **Engine** Unity · **Team** 5 · **For** Centennial College capstone

- Day/night cycle with skybox, exposure and light blending
- The whole night phase: shop with tabs, grid placement, move, rotate and sell
- Audio manager with fade-in/out jobs, and a data-driven cheat code system using UnityEvents
- Fixed AI code that overloaded the CPU (stutter, FPS drops and overheating) because objects weren't being disposed



VR · 4 MINIGAMES · SOLE PROGRAMMER

VR Carnival

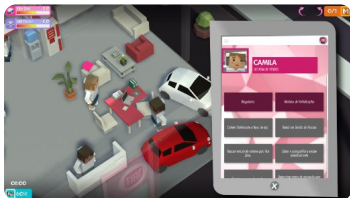
Four seated VR minigames in under a month · sole programmer

Role Sole Programmer · Game Designer · **Engine** Unity · SteamVR · **Team** 3 · **For** Centennial College VR course

- MiniGameManager singleton base class with overridable lifecycle hooks, so only one minigame runs at a time
- Dissolve shader effect for materializing the bow and hammer in the player's hand
- Extended SteamVR's interaction samples (grabbing, bow, buttons) into four distinct games

LEADERSHIP

- On a short-handed team, took on all the programming and scoped the project to four core games in under a month, polishing only after all four were built



TRAINING SIM · CLIENT: FIAT

Fiat Quality Manager Training

Data-driven training simulation for Fiat dealership Quality Managers

Role Dev Lead · Programmer · Game Design team · **Engine** Unity · **For** Fiat · D2R Studios

- Content pipeline: Fiat's Excel scenario sheets → JSON → NoSQL database, downloaded by Unity at runtime
- 76 real-world scenarios at launch, with a web admin so the client can add more without a new build

LEADERSHIP

- Dev lead and one of the main programmers, while also designing the game as part of a team where everyone helped shape it



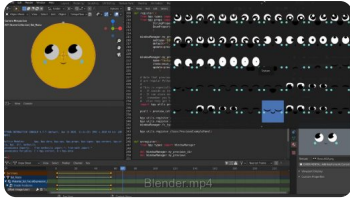
MULTI-SCREEN APP · CLIENT: VALE · STILL IN USE

Vale · Porto de Tubarão

Three-screen presentation system controlled from a tablet · 2012 → 2019

Role Desktop Developer (2012) · Rebuild Developer (2019) · **Year** 2012 / 2015 / 2019 · **Engine** Unity · **For** Vale · Zor Design · D2R Studios

- Custom sprite system written before Unity had native 2D
- Content pipeline through the Resources folder: new images only need the right file names, with no code or level changes
- In one month for the 2019 version: rebuilt the tablet app from scratch, moved the PC app off OnGUI and added real networking with UNet
- Still in use today, more than a decade after the first version



ANIMATION TOOL · PYTHON · STILL IN USE

Blender Expressions Add-on

Python add-on that speeds up facial-expression keyframing · built for D2R Studios' animator, 2019

Role Tool Designer · Programmer · **Year** 2019 · **Engine** Blender · Python · **For** D2R Studios

- Lists every facial texture for a character's face parts, read from a folder naming convention, so animators pick expressions from one panel
- Saves around 3 clicks per keyframe when swapping faces, plus a one-click keyframe button
- Built as a side project alongside my regular work, fitted into the studio's existing animation pipeline; the animator it was made for still uses it



HORROR · GAME JAM · TEAM LEAD

Coma · College Game Jam

Horror game jam on the theme "Another World" · led a team of newer developers

Role Team Lead · Programmer · **Year** 2021 · **Engine** Unity · **For** College midterm game jam

- Procedural maze generator (adapted from Kodeco's maze tutorial) whose output was saved and then decorated by hand
- Reusable trigger system built on UnityEvents, with cooldowns and one-shot options

LEADERSHIP

- Led a team of newer students (the coordinator placed me there to lead it) and set up a prefab workflow so the artists could do the level design themselves



EDUCATIONAL · LUA · UNITY

Mitologia da Programação

Educational game where players solve Greek-myth challenges by writing Lua

Role Game Designer · Level Designer · Programmer · **Engine** Unity · embedded Lua · **Team** 2

- Embedded a Lua interpreter in Unity, so players' code compiles and runs inside the game
- Checks each solution against the expected output, with custom hint messages or the compiler's own errors



PUZZLE PLATFORMER · SBGAMES AWARD · 2013

Loneliness

Shadow-puzzle platformer · SBGames 2013: 3rd in the Popular Vote + Best Short Paper

Role Game Designer · Level Designer · Programmer · **Year** 2013 · **For** PUC capstone

- Shadow replay system: moved from frame-data recording to input recording to fix desyncs on frame drops (the postmortem covers what that late rewrite cost)

MORE WORK

Animal Defense, Fiat Personalização, Custom Defense, Procedural Dungeon Generator, Chaveirinho, Digital Wonderland, Recifuturo. Details, videos and code at pedroloures.com/projects.